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```
clc;  
clear;  
close all;
```

DSSS SPREADING AND DESPREADING

```
% Data Bits  
data = [1 0 1 1 0];  
  
% Convert binary data (0,1) to bipolar form (-1,+1)  
data_bipolar = 2 * data - 1;  
  
% PN Sequence (Length = 5)  
pn = [1 -1 1 -1 1];
```

DSSS Spreading

```
% Spread each data bit using the PN sequence  
spread = kron(data_bipolar, pn);
```

DSSS Despreading

```
% Reshape spread signal into rows corresponding to PN sequence length  
rec = reshape(spread, length(pn), []);  
  
% Correlate each received chip group with PN sequence  
correlation = sum(rec .* pn');  
  
% Determine recovered bipolar data  
rec_data_bipolar = sign(correlation);  
  
% Convert bipolar data back to binary  
rec_data = (rec_data_bipolar + 1) / 2;
```

Display Results

```
disp('=====');
disp('      DSSS SPREAD SPECTRUM SYSTEM');
disp('=====');

disp('Original Data:');
disp(data);

disp('PN Sequence:');
disp(pn);

disp('Spread Signal:');
disp(spread);

disp('Correlation Output:');
disp(correlation);

disp('Recovered Data:');
disp(rec_data);
```

```
=====
      DSSS SPREAD SPECTRUM SYSTEM
=====
Original Data:
      1      0      1      1      0

PN Sequence:
      1     -1      1     -1      1

Spread Signal:
Columns 1 through 13
      1     -1      1     -1      1     -1      1     -1      1     -1      1     -1      1

Columns 14 through 25
     -1      1      1     -1      1     -1      1     -1      1     -1      1     -1

Correlation Output:
      5     -5      5      5     -5

Recovered Data:
      1      0      1      1      0
```

Verify Data Recovery

```
if isequal(data, rec_data)
    disp(' ');
    disp('Result: Original data successfully recovered.');
```

```
else
    disp(' ');
```

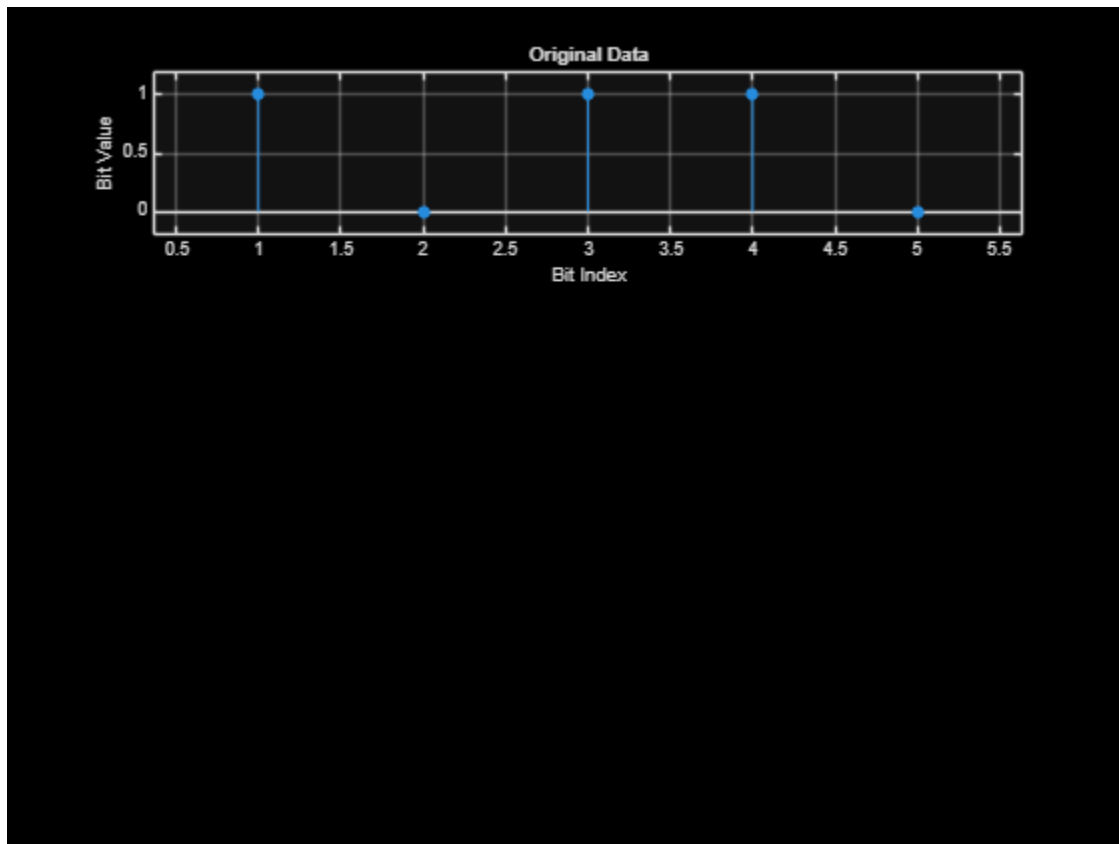
```
    disp('Result: Data recovery failed.');
```

```
end
```

Result: Original data successfully recovered.

Plot Original Data

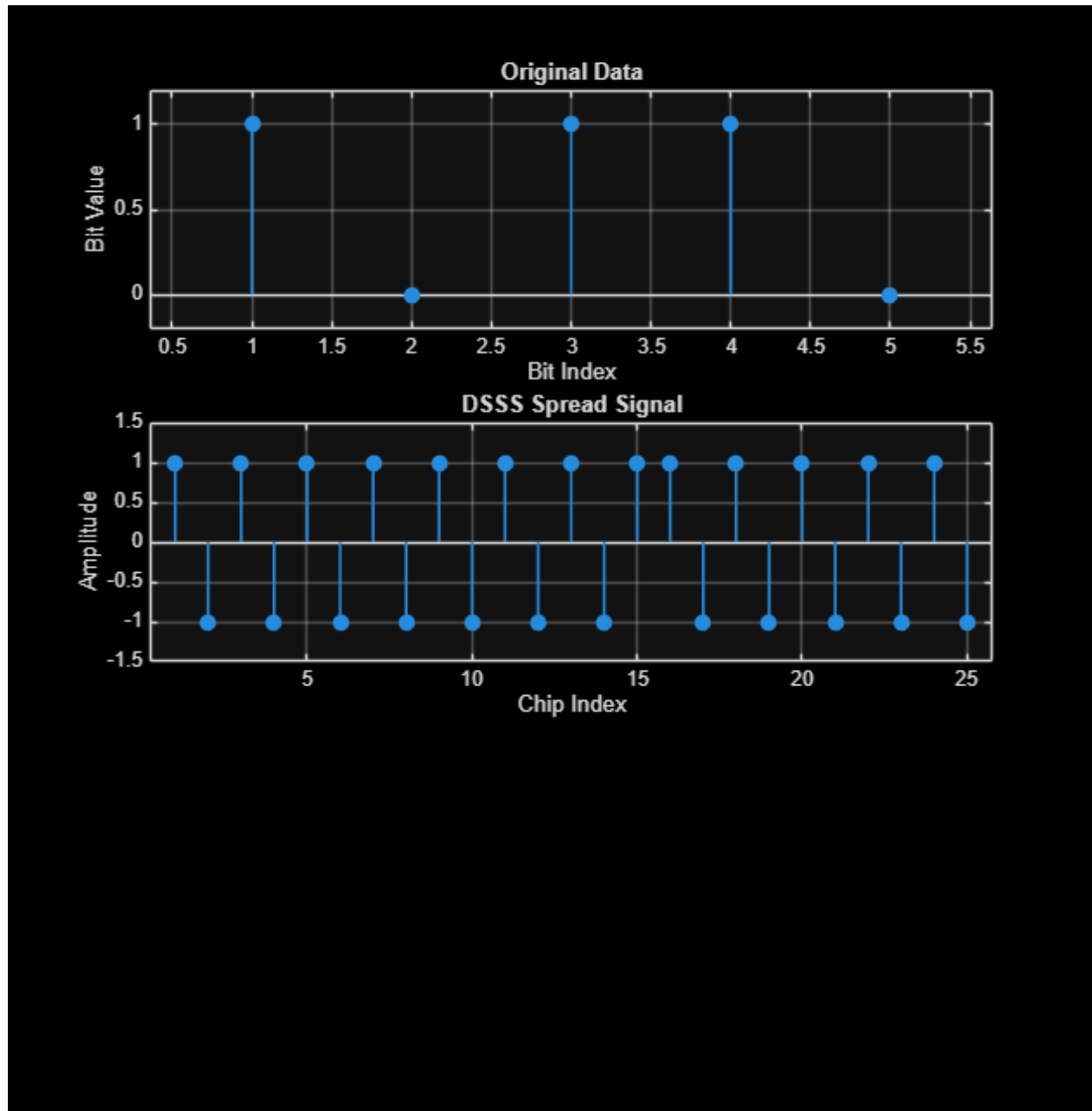
```
figure;  
  
subplot(3,1,1);  
stem(1:length(data), data, 'filled', 'LineWidth', 1.2);  
title('Original Data');  
xlabel('Bit Index');  
ylabel('Bit Value');  
ylim([-0.2 1.2]);  
grid on;
```



Plot DSSS Spread Signal

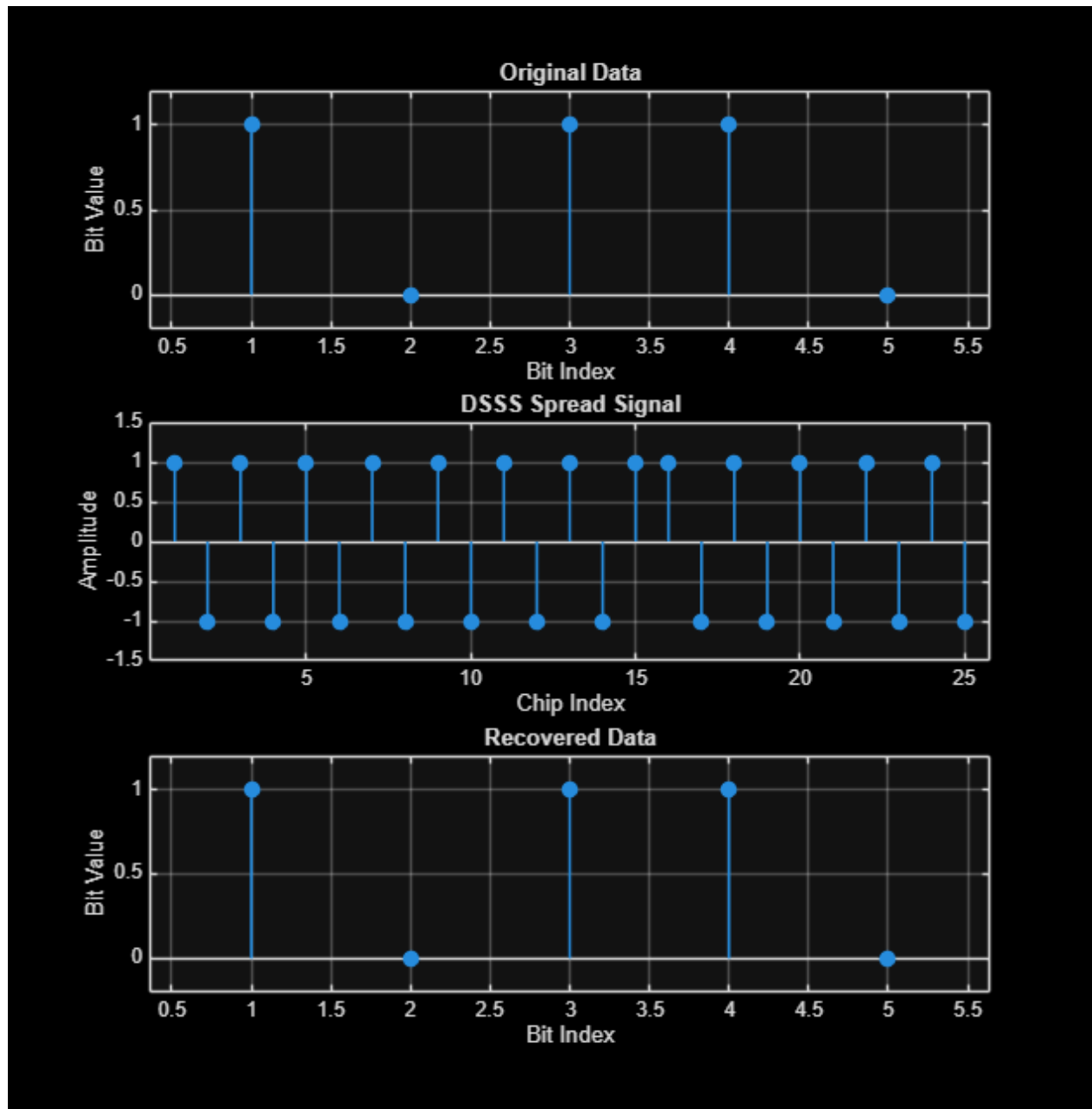
```
subplot(3,1,2);  
stem(1:length(spread), spread, 'filled', 'LineWidth', 1.2);  
title('DSSS Spread Signal');  
xlabel('Chip Index');  
ylabel('Amplitude');
```

```
ylim([-1.5 1.5]);
grid on;
```



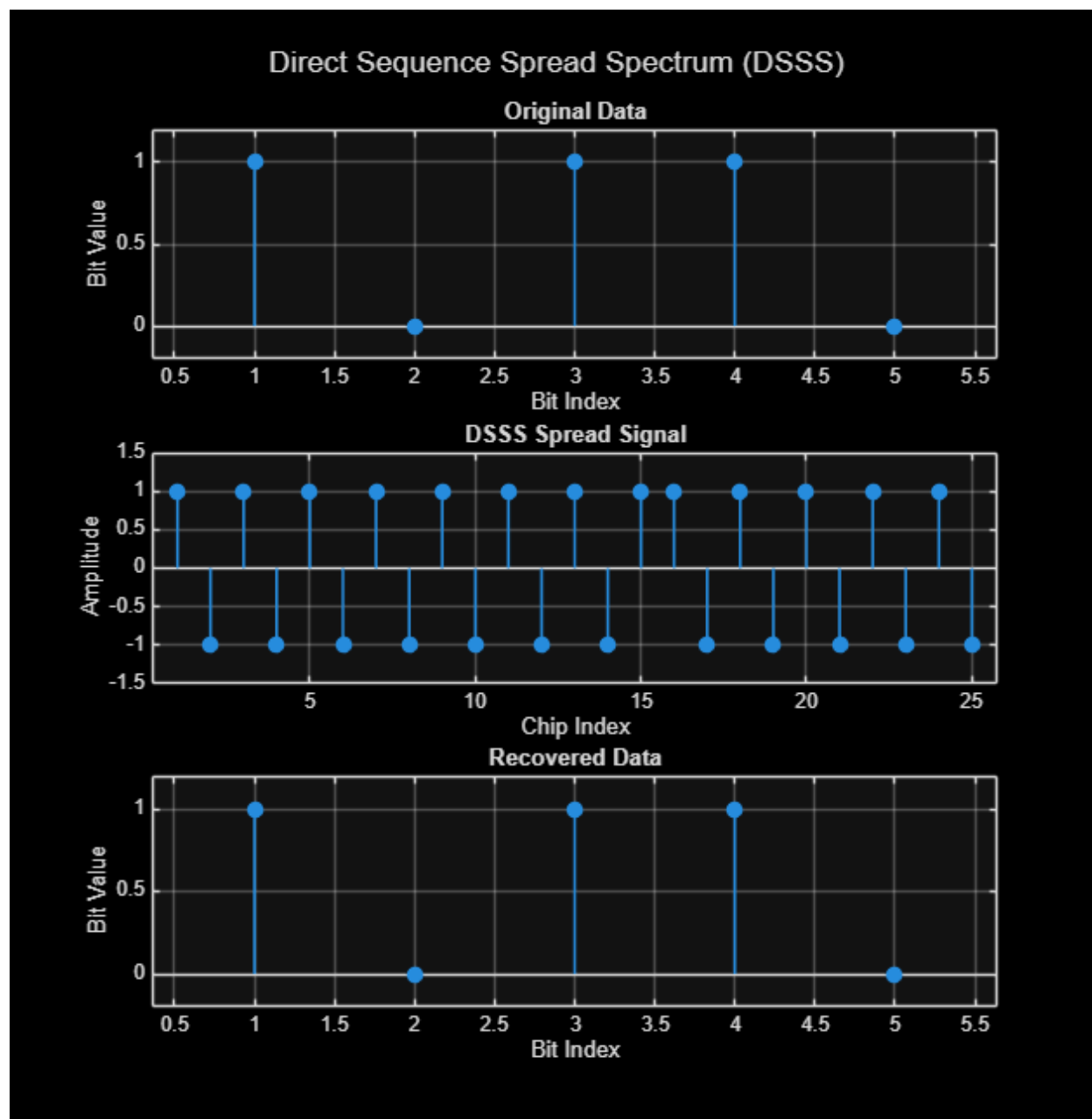
Plot Recovered Data

```
subplot(3,1,3);
stem(1:length(rec_data), rec_data, 'filled', 'LineWidth', 1.2);
title('Recovered Data');
xlabel('Bit Index');
ylabel('Bit Value');
ylim([-0.2 1.2]);
grid on;
```



Overall Figure Title

```
sgtitle('Direct Sequence Spread Spectrum (DSSS)');
```



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